PRACTICE SET

Absolute Value Inequalities: Special Cases — Set 1

Name: _____ Date: _____

Directions: Isolate the absolute value first if needed. Then choose the type and write the solution in interval notation. Let c be the value that makes the inside equal 0.

A	No solution	$\varnothing$	e.g. $\lvert x-3\rvert \le -2$ or $\lvert x-3\rvert < 0$
B	All real numbers	$(-\infty, \infty)$	e.g. $\lvert x-3\rvert > -2$ or $\lvert x-3\rvert \ge 0$
C	Only one solution	$\{c\}$	e.g. $\lvert x-3\rvert \le 0 \Rightarrow \{3\}$
D	All reals except one number	$(-\infty, c) \cup (c, \infty)$	e.g. $\lvert x-3\rvert > 0 \Rightarrow (-\infty, 3) \cup (3, \infty)$

1. $|-3x+5|>0$ ____ _____

2. $|x+6|>0$ ____ _____

3. $4\,|2x+9|-8<-16$ ____ _____

4. $|3x+3|<0$ ____ _____

5. $|x-1|\le 0$ ____ _____

6. $2\,|-2x-5|+6\le 6$ ____ _____

7. $|x-1|\ge -2$ ____ _____

8. $|x+4|\ge 0$ ____ _____

9. $5\,|-3x+3|-3\le -3$ ____ _____

10. $|5x+8|<-5$ ____ _____

11. $|x+8|>0$ ____ _____

12. $2\,|-3x-9|-7\le -11$ ____ _____

13. $|-2x+3|>0$ ____ _____

14. $|-x+6|\le 0$ ____ _____

15. $3\,|x+6|-3\le -3$ ____ _____

16. $|2x+9|>0$ ____ _____

17. $|5x-5|>0$ ____ _____

18. $4\,|x-5|+3\le 3$ ____ _____

19. $|4x-7|\le 0$ ____ _____

20. $|x+9|>0$ ____ _____

21. $5\,|-x+3|-7\ge -7$ ____ _____

22. $|2x-1|\le -6$ ____ _____

23. $|x-1|\le -3$ ____ _____

24. $5\,|2x+6|-5>-25$ ____ _____

25. $|-x+2|<-4$ ____ _____

26. $|x-1|<0$ ____ _____

27. $2\,|-3x-4|-8\ge -8$ ____ _____

28. $|-3x+4|\le 0$ ____ _____

29. $|-2x+6|>0$ ____ _____

30. $4\,|-3x-8|-1<-1$ ____ _____

31. $|2x+7|<0$ ____ _____

32. $|-3x-3|>0$ ____ _____

33. $4\,|-x+9|-8\ge -32$ ____ _____

34. $|x+1|\le -6$ ____ _____

35. $|-3x+3|\le 0$ ____ _____

36. $3\,|4x+4|-6>-9$ ____ _____

37. $|2x-5|>0$ ____ _____

38. $|-3x-6|\le 0$ ____ _____

39. $4\,|x+2|-8>-16$ ____ _____

40. $|5x-3|\ge 0$ ____ _____

41. $|-3x+3|>0$ ____ _____

42. $3\,|-x+9|+7\ge -5$ ____ _____

43. $|-x+9|\le 0$ ____ _____

44. $|3x+1|>0$ ____ _____

45. $3\,|-3x+1|+6<-9$ ____ _____

46. $|2x+6|\le 0$ ____ _____

47. $|2x-8|\ge -4$ ____ _____

48. $5\,|x+2|+7\le 7$ ____ _____

49. $|-3x-8|>-5$ ____ _____

50. $|-x-3|>0$ ____ _____

© 2026 FangMath. All rights reserved.